

# ELEMENTARY MENSURATION



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## ELEMENTARY MENSURATION

**Q 1.** Find the volume of cuboid 22 cm, by 12 cm, by 7.5 cm

- (a) 1980 cu cm
- (b) 1890 cu cm
- (c) 1680 cu cm
- (d) None of these

## ELEMENTARY MENSURATION

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## ELEMENTARY MENSURATION

**Q 2.** A wooden box of dimensions 8 m 7 m 6 m is to carry rectangular boxes of dimensions 8 cm 7 cm 6 cm. The maximum number of boxes that can be carried in the wooden box, is:

- (a) 9800000
- (b) 7500000
- (c) 1000000
- (d) 1200000

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- (a) 9800000
- (b) 7500000
- (c) 1000000**
- (d) 1200000

## ELEMENTARY MENSURATION

**Q3.** Find the weight (to the nearest kilogram) of an iron rod of square section, 10 metres long and 2.3 cm broad. A cubic cm of iron weighs 7.207 grams.

- (a) 40 kg
- (b) 39 kg
- (c) 38 kg
- (d) Data inadequate

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- (c) 38 kg**
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## ELEMENTARY MENSURATION

**Q 4.** The area of a side of a box is 120 sq cm. The area of the other side of the box is 72 sq cm. If the area of the upper surface of the box is 60 sq cm then find the volume of the box.

- (a)  $259200 \text{ cm}^3$
- (b)  $86400 \text{ cm}^3$
- (c)  $720 \text{ cm}^3$
- (d) Can't be determined



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- (b) 86400 cm<sup>3</sup>
- (c) 720 cm<sup>3</sup>**
- (d) Can't be determined

## ELEMENTARY MENSURATION

**Q 5.** The sum of length, breadth and height of a cuboid is 5 cm and its diagonal is 4 cm long. Find the total surface area of the cuboid.

- (a) 9 sq cm
- (b) 3 sq cm
- (c) 10 sq cm
- (d) Data inadequate

## ELEMENTARY MENSURATION

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- (a) 9 sq cm**
- (b) 3 sq cm
- (c) 10 sq cm
- (d) Data inadequate

## ELEMENTARY MENSURATION

**Q 6.** A cube has a diagonal 17.32 cm long. Find the volume of the cube.

- (a) 1000 cu cm
- (b) 1500 cu cm
- (c) 2000 cu cm
- (d) 2500 cu cm



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- (c) 2000 cu cm
- (d) 2500 cu cm

## ELEMENTARY MENSURATION

**Q 7.** How many cubic metres of earth must be dug out to sink a well 35 metres deep and 4 metres in diameter?

- (a) 220 cub m
- (b) 660 cub m
- (c) 440 cub m
- (d) Can't be determined

## ELEMENTARY MENSURATION

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- (c) 440 cub m**
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## ELEMENTARY MENSURATION

**Q 8.** The area of the curved surface of a cylinder is  $4400 \text{ cm}^2$  and the circumference of its base is  $110 \text{ cm}$ . Find the height and the volume of the cylinder.

- (a)  $40 \text{ cm}$ ,  $38500 \text{ cu cm}$
- (b)  $45 \text{ cm}$ ,  $38500 \text{ cu cm}$
- (c)  $40 \text{ cm}$ ,  $38500 \text{ cm}$
- (d)  $45 \text{ cm}$ ,  $38560 \text{ cu cm}$



## ELEMENTARY MENSURATION

**Q 8.** The area of the curved surface of a cylinder is  $4400 \text{ cm}^2$  and the circumference of its base is  $110 \text{ cm}$ . Find the height and the volume of the cylinder.

- (a) 40 cm, 38500 cu cm**
- (b) 45 cm, 38500 cu cm**
- (c) 40 cm, 38500 cm**
- (d) 45 cm, 38560 cu cm**

## ELEMENTARY MENSURATION

**Q 9.** The largest sphere is carved out of a cube of side 7 cm. Find the volume of the sphere (Take  $\pi = 3.14$ ).

- (a) 179.6 cu cm
- (b) 180.6 cu cm
- (c) 176.9 cu cm
- (b) 189.6 cu cm

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## ELEMENTARY MENSURATION

**Q 10.** Find the weight of a iron shell, the external and internal diameters of which are 13 cm and 10 cm respectively, if 1 cu cm of iron weighs 8 gms.

- (a) 6 kg
- (b) 6.015 kg
- (c) 5.016 kg
- (d) 5.015 kg

## ELEMENTARY MENSURATION

**Q 10.** Find the weight of a iron shell, the external and internal diameters of which are 13 cm and 10 cm respectively, if 1 cu cm of iron weighs 8 gms.

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- (d) 5.015 kg

## ELEMENTARY MENSURATION

**Q 11.** How many bullets can be made out of a cube of lead whose edge measures 22 cm, each bullet being 2 cm in diameter?

- (a) 5324
- (b) 2662
- (c) 1347
- (d) 2541

## ELEMENTARY MENSURATION

**Q 11.** How many bullets can be made out of a cube of lead whose edge measures 22 cm, each bullet being 2 cm in diameter?

- (a) 5324
- (b) 2662
- (c) 1347
- (d) 2541**

## ELEMENTARY MENSURATION

**Q 12.** Find the curved surface area of a hemisphere of radius 28 cm.

- (a) 4928 sq cm
- (b) 4298 sq cm
- (c) 4982 sq cm
- (d) None of these

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## ELEMENTARY MENSURATION

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## ELEMENTARY MENSURATION

**Q 13.** The height of a cone is 16 cm and the diameter of its base is 24 cm. Find its slant height.

- (a) 10 cm
- (b) 20 cm
- (c) 15 cm
- (d) 25 cm

## ELEMENTARY MENSURATION

**Q 13.** The height of a cone is 16 cm and the diameter of its base is 24 cm. Find its slant height.

(a) 10 cm

**(b) 20 cm**

(c) 15 cm

(d) 25 cm

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## ELEMENTARY MENSURATION

**Q 14.** The diameter of base of a right circular cone is 6 cm and its perpendicular height is 3 cm. Find the slant height of the cone.

- (a) 6 cm
- (b) none of these
- (c)  $6\sqrt{3}$  cm
- (d)  $\sqrt{63}$  cm

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## ELEMENTARY MENSURATION

**Q 14.** The diameter of base of a right circular cone is 6 cm and its perpendicular height is 3 cm. Find the slant height of the cone.

(a) 6 cm

**(b) none of these**

(c)  $6\sqrt{3}$  cm

(d)  $\sqrt{63}$  cm

## ELEMENTARY MENSURATION

**Q 15.** A cone of height 7 cm and base radius 3 cm is carved from a rectangular block of wood 10 cm 5 cm 2 cm. Calculate the percentage of wood wasted.

- (a) 66%
- (b) 37%
- (c) 67%
- (d) 34%

## ELEMENTARY MENSURATION

**Q 15.** A cone of height 7 cm and base radius 3 cm is carved from a rectangular block of wood 10 cm 5 cm 2 cm. Calculate the percentage of wood wasted.

(a) 66%

(b) 37%

(c) 67%

**(d) 34%**

## ELEMENTARY MENSURATION

**Q 16.** Radius of the base of a right circular cone is 30 cm and the height of the cone is 40 cm. Find the total surface area of the cone.

- (a)  $240 \pi$  sq cm
- (b)  $2400 \pi$  sq cm
- (c)  $1600 \pi$  sq cm
- (d)  $1680 \pi$  sq cm



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- (b)  $2400 \pi$  sq cm**
- (c)  $1600 \pi$  sq cm
- (d)  $1680 \pi$  sq cm

## ELEMENTARY MENSURATION

**Q 17.** The circumference of one end of a frustum of a right circular cone is 48 cm and of the other end 34 cm, the height of the frustum is 10 cm, find its volume.

- (a) 1250 cub cm approx
- (b) 1850 cub cm approx
- (c) 1350 cub cm approx
- (d) 1360 cub cm approx

## ELEMENTARY MENSURATION

**Q 17.** The circumference of one end of a frustum of a right circular cone is 48 cm and of the other end 34 cm, the height of the frustum is 10 cm, find its volume.

- (a) 1250 cub cm approx
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- (c) 1350 cub cm approx**
- (d) 1360 cub cm approx

## ELEMENTARY MENSURATION

**Q 18.** The surface of a cube is  $1176 \text{ cm}^2$  . The volume of this cube is

- (a)  $7056 \text{ cm}^3$
- (b)  $4704 \text{ cm}^3$
- (c)  $2744 \text{ cm}^3$
- (d)  $3528 \text{ cm}^3$

## ELEMENTARY MENSURATION

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- (c)  $2744 \text{ cm}^3$**
- (d)  $3528 \text{ cm}^3$

## ELEMENTARY MENSURATION

**Q 19.** Three cubes of metal whose edges are 30, 40 and 50 cm respectively are melted and formed into a single cube. If there be no loss of metal in the process find the side of the new cube.

- (a) 60 cm
- (b) 64 cm
- (c) 90 cm
- (d) 80 cm

## ELEMENTARY MENSURATION

**Q 19.** Three cubes of metal whose edges are 30, 40 and 50 cm respectively are melted and formed into a single cube. If there be no loss of metal in the process find the side of the new cube.

**(a) 60 cm**

(b) 64 cm

(c) 90 cm

(d) 80 cm

## ELEMENTARY MENSURATION

**Q 20.** A cube of sides 6 cm is melted and smaller cubes of sides 3 cm each are formed. How many such cubes are possible?

- (a) 16
- (b) 8
- (c) 27
- (d) Data inadequate



## ELEMENTARY MENSURATION

**Q 20.** A cube of sides 6 cm is melted and smaller cubes of sides 3 cm each are formed. How many such cubes are possible?

- (a) 16
- (b) 8**
- (c) 27
- (d) Data inadequate

## ELEMENTARY MENSURATION

**Q 21.** The internal diameter of an iron pipe is 6 cm and the length is 2.8 metres. If the thickness of the metal be 5 mm and 1 cu cm of iron weighs 8 gm, find the weight of the pipe.

- (a) 288.2 kg
- (b) 22.88 kg
- (c) 822.2 kg
- (d) None of these

## ELEMENTARY MENSURATION

**Q 21.** The internal diameter of an iron pipe is 6 cm and the length is 2.8 metres. If the thickness of the metal be 5 mm and 1 cu cm of iron weighs 8 gm, find the weight of the pipe.

- (a) 288.2 kg
- (b) 22.88 kg**
- (c) 822.2 kg
- (d) None of these

## ELEMENTARY MENSURATION

**Q 22.** A copper sphere of diameter 12 cm is drawn into a wire of diameter 2 cm. Find the length of the wire.

- (a) 288 cm
- (b) 284 cm
- (c) 286 cm
- (d) None of these

## ELEMENTARY MENSURATION

**Q 22.** A copper sphere of diameter 12 cm is drawn into a wire of diameter 2 cm.  
Find the length of the wire.

- (a) 288 cm**
- (b) 284 cm
- (c) 286 cm
- (d) None of these

## ELEMENTARY MENSURATION

**Q 23.** A cylinder of radius 6 cm and height 8 cm is melted and the same mass is used to create a sphere. What will be the radius of the sphere?

- (a) 7 cm
- (b) 4 cm
- (c) 6 cm
- (d)  $\sqrt[3]{36}$  cm

## ELEMENTARY MENSURATION

**Q 23.** A cylinder of radius 6 cm and height 8 cm is melted and the same mass is used to create a sphere. What will be the radius of the sphere?

- (a) 7 cm
- (b) 4 cm
- (c) 6 cm**
- (d)  $\sqrt[3]{36}$  cm

## ELEMENTARY MENSURATION

**Q 24.** A cylinder of length 1 metre and diameter 15 cm is melted down and cast into spheres of diameter 5 cm. How many spheres can be made?

- (a) 270
- (b) 260
- (c) 290
- (d) 370



## ELEMENTARY MENSURATION

**Q 24.** A cylinder of length 1 metre and diameter 15 cm is melted down and cast into spheres of diameter 5 cm. How many spheres can be made?

**(a) 270**

(b) 260

(c) 290

(d) 370

## ELEMENTARY MENSURATION

**Q 25.** There is a cone of radius 18 cm and height 24 cm. Find the radius of the greatest sphere that can be carved out of that cone.

- (a) 9 cm
- (b) 12 cm
- (c) 6 cm
- (d) Can't be determined

## ELEMENTARY MENSURATION

**Q 25.** There is a cone of radius 18 cm and height 24 cm. Find the radius of the greatest sphere that can be carved out of that cone.

- (a) 9 cm
- (b) 12 cm
- (c) 6 cm
- (d) Can't be determined

## ELEMENTARY MENSURATION

**Q 26.** The curved surface areas of two spheres are in the ratio 4 : 5. Find the ratio of their volumes.

- (a) 8 : 5
- (b)  $8 : 5\sqrt{3}$
- (c)  $8 : 5\sqrt{5}$
- (d)  $5\sqrt{5} : 8$

## ELEMENTARY MENSURATION

**Q 26.** The curved surface areas of two spheres are in the ratio 4 : 5. Find the ratio of their volumes.

(a)  $8 : 5$

(b)  $8 : 5\sqrt{3}$

(c)  $8 : 5\sqrt{5}$

(d)  $5\sqrt{5} : 8$

## ELEMENTARY MENSURATION

**Q 27.** If the radius of a cylinder becomes 3 times and the height  $\frac{1}{3}$  times, what is the ratio between the new curved surface area and the previous curved surface area of the cylinder?

- (a) 1 : 1
- (b) 1 : 2
- (c) 2 : 1
- (d) Can't be determined

## ELEMENTARY MENSURATION

**Q 27.** If the radius of a cylinder becomes 3 times and the height  $\frac{1}{3}$  times, what is the ratio between the new curved surface area and the previous curved surface area of the cylinder?

- (a) 1 : 1**
- (b) 1 : 2**
- (c) 2 : 1**
- (d) Can't be determined**

## ELEMENTARY MENSURATION

**Q 28.** A right-angled triangle having base 6 metres and height equal to 8 metres, is turned around the height. Find the volume of the cone thus formed. Also find the surface area.

- (a)  $96 \pi$  ,  $60 \pi$
- (b)  $48 \pi$  ,  $60 \pi$
- (c)  $96 \pi$  ,  $120 \pi$
- (d)  $48 \pi$  ,  $30 \pi$



## ELEMENTARY MENSURATION

**Q 28.** A right-angled triangle having base 6 metres and height equal to 8 metres, is turned around the height. Find the volume of the cone thus formed. Also find the surface area.

- (a)  $96 \pi$  ,  $60 \pi$
- (b)  $48 \pi$  ,  $60 \pi$
- (c)  $96 \pi$  ,  $120 \pi$
- (d)  $48 \pi$  ,  $30 \pi$

## ELEMENTARY MENSURATION

**Q 29.** The length, breadth and height of a cuboid are made 2, 3 and 4 times respectively. Find the percentage increase in its volume.

- (a) 230%
- (b) 23%
- (c) 2300%
- (d) data inadequate

## ELEMENTARY MENSURATION

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(a) 230%

(b) 23%

**(c) 2300%**

(d) data inadequate

## ELEMENTARY MENSURATION

**Q 30.** If only length of a cuboid is made 4 times then, find the percentage increase in its volume.

- (a) 6300%
- (b) 640%
- (c) 300%
- (d) None of these

## ELEMENTARY MENSURATION

**Q 30.** If only length of a cuboid is made 4 times then, find the percentage increase in its volume.

- (a) 6300%
- (b) 640%
- (c) 300%**
- (d) None of these

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## ELEMENTARY MENSURATION

**Q 31.** Each edge of a cube is increased by 10%. What is the percentage increase in its volume.

- (a) 20%
- (b) 33.1%
- (c) 20.5%
- (d) 22%

## ELEMENTARY MENSURATION

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- (a) 20%
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# THANK YOU

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